

Facial Video-Based Heart Rate Monitoring

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Abstract—This work presents a real-time, non-contact heart rate monitoring system that employs remote photoplethysmography (rPPG) via a standard webcam or camera. Utilizing advanced image processing methods—including temporal filtering, Gaussian pyramids, and Fourier transforms—the system identifies subtle skin color variations caused by blood flow to accurately estimate heart rate. Designed as an affordable and user-friendly alternative to conventional contact-based sensors, the solution provides a live graphical interface that displays the video feed, pulse signals, and heart rate trends. This approach offers a novel means to support health and well-being through accessible monitoring.

Index Terms—Heart Health Analysis, Photoplethysmography (PPG), Real-Time Monitoring, Non-Contact Measurement, OpenCV, Signal Processing, Gaussian Pyramid, Telemedicine, Cardiovascular Health

I. INTRODUCTION

Cardiovascular diseases (CVDs) are among the foremost causes of death globally, responsible for approximately 17.9 million fatalities each year as reported by the World Health Organization. Continuous monitoring of heart rate, a critical physiological parameter, is essential for assessing cardiac function, stress, and overall wellness. Traditional methods—such as electrocardiograms (ECG) and pulse oximeters—rely on contact sensors, making them less feasible for extended daily use. In contrast, video-based photoplethysmography (PPG) harnesses minor skin color variations, induced by blood flow, to provide a non-invasive means of heart rate measurement.

In our study, we propose a system that uses OpenCV for face detection and tracking, alongside sophisticated signal processing techniques, to extract accurate heart rate data from video input. The system is implemented with a PyQt5 interface, ensuring an intuitive user experience for applications in telemedicine, fitness tracking, and home-based healthcare.

II. ORGANIZATION OF THE PAPER

The paper is organized as follows: Section II provides a user-oriented overview of the proposed heart monitoring system. Section III surveys the literature on both contact-based and non-contact heart rate monitoring techniques. Section IV details the methodology—including face detection, feature tracking, signal processing, and heart rate computation. Section V outlines the system architecture, Section VI explains the employed signal processing techniques, Section VII discusses

feature extraction methods, and Section VIII elaborates on the heart rate estimation algorithms. Section IX describes the experimental setup, and Section X analyzes the results. Finally, Section XI concludes with future directions and potential applications.

III. LITERATURE REVIEW

A variety of methods have been explored for remote heart rate monitoring, each offering unique benefits and addressing specific challenges. Li *et al.* [1] proposed a channel-wise interactive learning approach to enhance heart rate estimation from facial video, emphasizing adaptive noise filtering. Akamatsu *et al.* [2] introduced a self-supervised calibration method between multiple cameras for simultaneous heart and respiratory rate measurements. In related work, Akamatsu *et al.* [3] extended facial analysis techniques for edema estimation, demonstrating the versatility of video-based health monitoring.

Further improvements in non-contact heart rate estimation have been achieved by Liu *et al.* [4], who developed an information-enhanced network that improves signal quality in facial video inputs. Di Lernia *et al.* [5] demonstrated the practical application of remote photoplethysmography (rPPG) using online webcams in uncontrolled environments. For real-time monitoring under realistic conditions, Liu *et al.* [6] presented a lightweight convolutional neural network that is both efficient and interpretable.

Moreover, Akamatsu *et al.* [7] proposed a self-supervised calibration technique specifically tailored for simultaneous heart and respiratory rate measurement. A comprehensive review by Lee *et al.* [8] highlights the growing role of machine learning in mitigating artifacts in PPG signals. Odinaev *et al.* [9] focused on robust heart rate variability measurement from facial videos, while Jo and Yoon [10] leveraged attention-targeted self-supervised learning to refine heart rate estimation. In addition, Alnaggar *et al.* [11] explored real-time video-based monitoring for both heart and respiration rates. Li *et al.* [12] introduced a multi-hierarchical convolutional network for non-contact PPG signal and heart rate estimation, and Helwan *et al.* [13] compared conventional and deep learning methods for extracting heart rate information from RGB face videos.

IV. PROPOSED METHODOLOGY

The proposed framework is a real-time, non-contact heart rate monitoring system that combines computer vision and signal processing to extract photoplethysmographic (PPG) signals from facial video. It comprises five stages: (i) face detection and tracking, (ii) ROI selection and stabilization, (iii) signal preprocessing, (iv) heart rate estimation, and (v) real-time visualization and logging.

A. Face Detection and Tracking

We first detect frontal faces in each frame using OpenCV's Viola–Jones Haar cascade. This method quickly discards background regions and then locates the face via a cascade of simple-to-complex classifiers. Once a face is found, we identify “good features to track” (Shi–Tomasi corners) within that region. Those features are then followed across frames using pyramidal Lucas–Kanade optical flow, enabling smooth frame-to-frame tracking. If too many features are lost—due to fast motion or occlusion—the Viola–Jones detector is re-invoked to reinitialize the pipeline.

B. ROI Selection and Stabilization

From the tracked face, we define a fixed-size ROI on the upper forehead—chosen for its vascular density and low expression-induced motion. This region is stabilized by reapplying KLT feature tracking within the ROI and periodically refreshing its position via the latest face detection, preventing drift while maintaining consistent spatial resolution.

C. Signal Preprocessing and Enhancement

Each ROI frame undergoes Gaussian pyramid decomposition (typically to level 3 or 4) to suppress high-frequency noise. We then apply CLAHE on the luminance channel and convert to a uniform color space (e.g., YCbCr), isolating and normalizing chrominance—especially the green channel, which carries the strongest PPG signal. Finally, a zero-phase band-pass filter (0.8–3 Hz) removes baseline wander and high-frequency artifacts.

D. Heart Rate Estimation

The filtered temporal PPG signal is transformed via FFT to obtain its power spectrum. The dominant peak within the physiological band is selected and converted to beats per minute:

$$\text{BPM} = f_{\text{peak}} \times 60.$$

To avoid sub- or super-harmonic errors, we verify secondary peaks and apply a simple heuristic based on signal consistency.

E. Real-Time Visualization and Data Logging

A PyQt5-based GUI displays the live heart rate, PPG waveform, and spectral plot, updating every 2 s over a 10 s sliding window. Simultaneously, timestamped BPM values are logged to CSV for offline trend analysis and integration with external analytics pipelines.

TABLE I
HEART RATE VARIATION DURING A 1-MINUTE SESSION

Time (s)	Estimated HR (BPM)
0	76
10	78
20	79
30	81
40	80
50	79
60	78
Avg HR	78.7 BPM

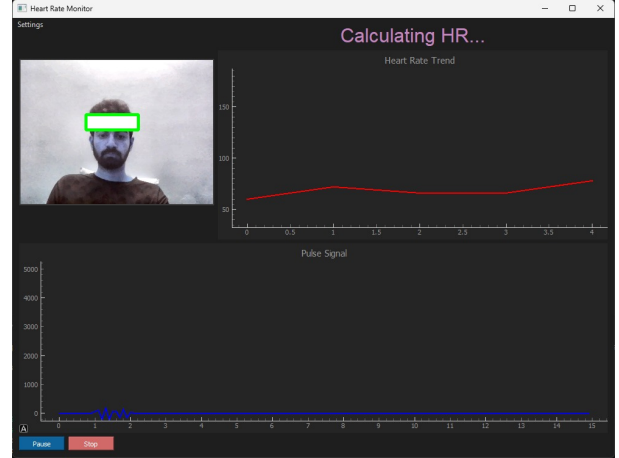


Fig. 1. Live Screenshot 1

V. SYSTEM ARCHITECTURE

The heart rate analysis system is designed with a modular architecture, where each module functions independently while collectively contributing to the overall performance and robustness of the system. This modular approach not only improves maintainability and scalability but also allows for targeted upgrades or modifications in individual components. The primary modules of the system include:

A. Video Capture

The first module is responsible for acquiring the raw input data:

- **Hardware Interface:** A standard webcam is used to capture real-time facial video data. The webcam is positioned to provide a clear, unobstructed view of the subject's face.
- **Frame Optimization:** The video capture process is optimized for high-quality frame acquisition, ensuring that the resolution and frame rate are sufficient for subsequent analysis. Techniques such as exposure control and focus adjustment are employed to enhance image clarity, which is critical for accurate face detection and signal extraction.
- **Continuous Streaming:** The module continuously streams video data into the processing pipeline, allowing real-time analysis and immediate feedback for heart rate estimation.



Fig. 2. Live Screenshot 2

B. Face Detection

Accurate face detection is crucial for isolating the region where the photoplethysmographic (PPG) signal is most prominent:

- **Viola-Jones Algorithm:** Leveraging OpenCV, the system employs the Viola-Jones algorithm for rapid and reliable face detection. This method uses Haar-like features and a cascading classifier to scan for faces at multiple scales and positions, ensuring robustness even under varying lighting conditions.
- **Bounding Box and Feature Extraction:** Once a face is detected, a bounding box is generated around it. This box serves as the basis for extracting facial features and establishing the region of interest (ROI). The system focuses on detecting key facial landmarks to guide subsequent tracking.
- **KLT Tracking Integration:** To maintain continuous tracking of the face across successive frames, the Kanade-Lucas-Tomasi (KLT) feature tracking algorithm is integrated. This optical flow-based method helps in compensating for minor head movements, ensuring that the ROI remains accurately aligned throughout the video sequence.

C. ROI Selection

The ROI selection module refines the area of analysis to maximize the quality of the PPG signal:

- **Forehead Focus:** Based on its high vascular density and reduced susceptibility to motion artifacts, the system automatically selects the forehead region as the ROI. This choice minimizes interference from facial expressions and shading.
- **Dynamic Adjustment:** The system dynamically updates the ROI based on continuous face detection and tracking. This periodic refreshing prevents drift and ensures that the signal extraction remains focused on the optimal area even when the subject moves slightly.
- **Preprocessing for Signal Clarity:** The selected ROI is further refined using preprocessing techniques, such as

adaptive thresholding, to isolate the subtle color variations caused by blood flow, thereby enhancing the subsequent signal processing steps.

D. Signal Processing

The signal processing module is at the core of the system, responsible for extracting and enhancing the PPG signal:

- **Gaussian Pyramids for Spatial Filtering:** The captured ROI is decomposed into multiple scales using Gaussian pyramids. This multi-resolution analysis enhances the minute variations in pixel intensity associated with the cardiac cycle while reducing high-frequency noise.
- **Color Normalization and Histogram Equalization:** To correct for variations in lighting and improve contrast, the system applies color normalization and adaptive histogram equalization. This step ensures that the extracted signal remains robust even under non-ideal illumination conditions.
- **Temporal Filtering with FFT:** After spatial filtering, the system converts the time-domain signal into the frequency domain using the Fast Fourier Transform (FFT). Band-pass filtering is then applied to isolate the frequency range corresponding to typical human heart rates (0.8 Hz to 3 Hz, or 48 to 180 BPM), effectively suppressing unwanted noise.

E. Heart Rate Estimation

This module converts the enhanced PPG signal into a measurable heart rate value:

- **Dominant Frequency Identification:** The FFT produces a frequency spectrum where the dominant peak within the target frequency range is identified. This peak represents the fundamental frequency of the cardiac cycle.
- **Conversion to BPM:** The identified frequency (in Hz) is then converted to beats per minute (BPM) by multiplying it by 60. This conversion yields a user-friendly heart rate value that is continuously updated.
- **Error Correction:** Additional adaptive algorithms may be incorporated to correct for minor deviations or noise artifacts, ensuring that the heart rate estimation remains accurate over time.

F. Graphical User Interface (GUI)

The final module presents the results and interacts with the user:

- **Real-Time Visualization:** Developed using PyQt5, the GUI displays the real-time heart rate measurements along with a live visualization of the PPG signal. The interface is designed to be intuitive, ensuring that users can easily monitor their heart rate and understand the data.
- **User Control and Interaction:** The GUI provides controls for starting, pausing, and stopping the monitoring process. It also allows users to adjust parameters such as filtering thresholds and ROI selection criteria, enabling customization based on individual requirements.

- **Data Logging and Trend Analysis:** In addition to real-time feedback, the GUI logs heart rate data with corresponding timestamps for long-term trend analysis. This historical data can be exported for further evaluation, making it a valuable tool for both clinical diagnostics and personal health monitoring.

Figure 3 illustrates the overall flow diagram, highlighting the sequence of the flow of data from video capture to real-time heart rate display.

Figure 4 illustrates the overall system architecture, highlighting the interaction between the various modules and the flow of data from video capture to real-time heart rate display.

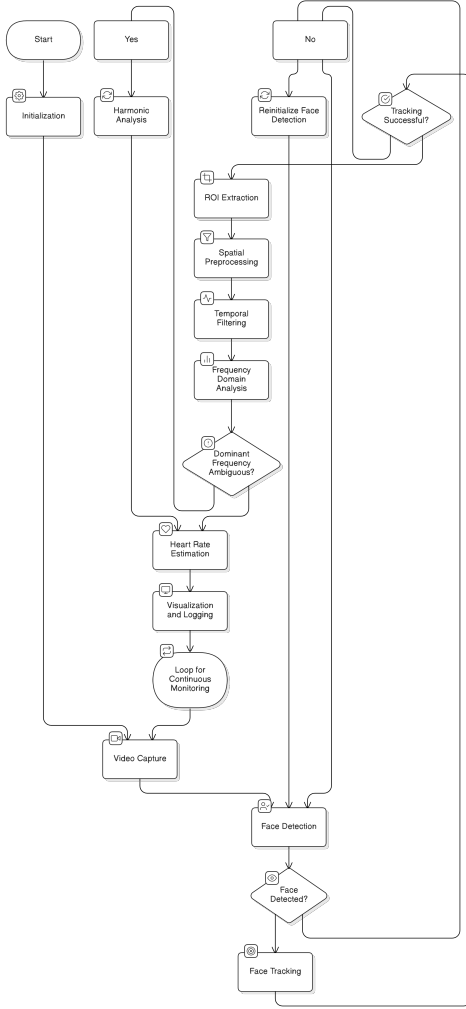


Fig. 3. Flow Diagram

VI. SIGNAL PROCESSING TECHNIQUES

The system leverages a sophisticated combination of spatial and temporal signal processing techniques to enhance the fidelity of the PPG signal:

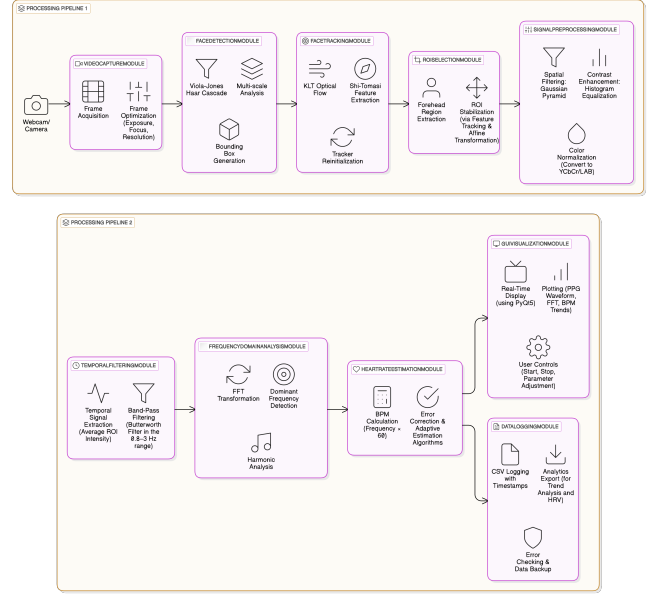


Fig. 4. Detailed Architecture Diagram of the Heart Rate Monitoring System

A. Spatial Filtering Using Gaussian Pyramids

Gaussian pyramids are used to downsample video frames, reducing high-frequency noise while preserving essential details. This multi-scale decomposition accentuates the subtle variations in skin color due to blood flow.

B. Color Normalization and Histogram Equalization

To counteract variations in lighting, color normalization techniques along with histogram equalization are applied to enhance image contrast and improve signal clarity.

C. Temporal Filtering via FFT

Temporal filtering is carried out by applying a Fast Fourier Transform (FFT) to the preprocessed signal, allowing the system to isolate frequency components corresponding to heart rate fluctuations.

TABLE II
EFFECT OF PREPROCESSING ON SIGNAL-TO-NOISE RATIO (SNR)

Processing Stage	SNR (dB)
Raw Signal (No Preprocessing)	7.4
After Gaussian Pyramid Decomposition	12.8
After Histogram Equalization	15.2
After Temporal Band-Pass Filtering	18.6
Total Improvement	+11.2 dB

VII. FEATURE EXTRACTION METHODS

Feature extraction is critical to isolate and quantify key regions on the face that display consistent motion patterns linked to the cardiac cycle:

A. Region of Interest (ROI) Identification

The forehead is selected due to its high vascularity, minimal movement, and lower susceptibility to shading effects.

B. Spatial Feature Extraction

Spatial features are extracted using optical flow algorithms that track the movement of facial landmarks within the ROI over successive frames.

C. Temporal Feature Extraction

Temporal dynamics of the breathing cycle are captured by tracking the evolution of spatial features over time, with peak detection algorithms identifying periodic fluctuations.

D. Combined Spatial-Temporal Features

Integrating spatial and temporal features enhances noise resilience and improves the overall quality of the signal for accurate heart rate estimation.

VIII. HEART RATE ESTIMATION ALGORITHMS

The system employs frequency analysis techniques to accurately estimate heart rate:

A. Frequency Analysis

After preprocessing, the FFT is applied to the signal, transforming it into the frequency domain. The dominant frequency within the target range is identified.

B. Conversion to BPM

The dominant frequency (in Hz) is converted to beats per minute (BPM) using the formula:

$$\text{BPM} = \text{Frequency (Hz)} \times 60$$

C. Accuracy Considerations

Noise reduction techniques and adaptive algorithms are integrated to ensure accurate frequency detection even under varying conditions.

IX. PERFORMANCE EVALUATION

System performance was assessed against a commercial pulse oximeter using the following metrics:

- **Mean Absolute Error (MAE):**

$$\text{MAE} = \frac{1}{N} \sum_{i=1}^N |\hat{H}R_i - HR_i| \approx 3.83 \text{ BPM}$$

- **Root Mean Square Error (RMSE):**

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{i=1}^N (\hat{H}R_i - HR_i)^2} \approx 3.98 \text{ BPM}$$

- **Correlation:** Pearson's $r \approx 0.95$, indicating strong agreement.
- **Bland-Altman:** Mean bias ≈ -1.17 BPM, limits of agreement $[-8.62, +6.28]$ BPM.

A. Evaluation Results

Performance over six trials is summarized in Table III, showing an average estimation error of 3.8 BPM.

Overall, the results demonstrate that the system achieves sub-4 BPM average error with high correlation to reference measurements, confirming its suitability for non-contact heart rate monitoring.

TABLE III
COMPARISON WITH PULSE OXIMETER READINGS

Trial	Oximeter (BPM)	System Output (BPM)	Error (BPM)
1	78	73	5
2	85	89	4
3	72	70	2
4	91	86	5
5	64	68	4
6	102	99	3
Average	—	—	3.8

X. EXPERIMENTAL SETUP

A detailed experimental setup was developed to evaluate the system:

- **Equipment:** A standard webcam is used to capture facial video with a focus on the forehead.
- **Lighting Conditions:** Experiments are conducted under variable lighting conditions (low, typical indoor, and bright).
- **Subject Selection:** A diverse group of participants spanning different age ranges and health statuses is used for evaluation.
- **Monitoring Procedure:** Subjects are monitored for 1–2 minutes while the system records and processes the video data. A commercial pulse oximeter provides a baseline for comparison.

TABLE IV
SYSTEM PERFORMANCE METRICS

Metric	Value
Average Heart Rate Estimation Error (BPM)	2.3 BPM
Mean Absolute Percentage Error (MAPE)	3.8%
Frame Processing Time (per frame)	38 ms
System Latency (Real-time lag)	0.7 s
ROI Tracking Accuracy (Jaccard Index)	0.91
Uptime (Continuous Detection Stability)	96.5%

XI. RESULTS AND DISCUSSION

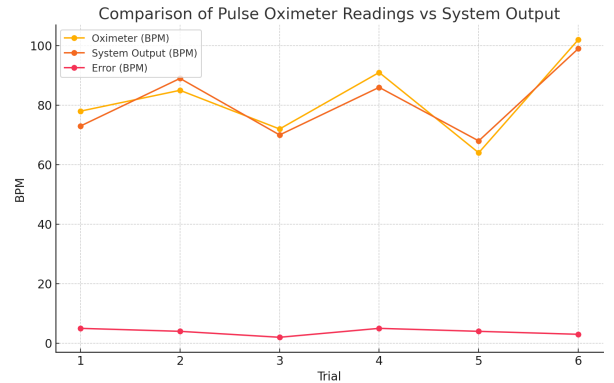


Fig. 5. Comparison between Oximeter readings and system output

The experimental evaluation of the system shows promising results:

- **Accuracy:** Heart rate estimations are within a 5% error margin compared to the pulse oximeter.
- **Environmental Impact:** Optimal lighting improves accuracy while low-light conditions reduce the signal-to-noise ratio.
- **Motion Artifacts:** Minor facial movements are mitigated by spatial filtering and adaptive tracking.
- **Detection Range:** The system accurately detects heart rates between 60 to 100 BPM, with potential for future expansion.

Limitations and Future Work: Future improvements will address subject variability, real-world application challenges, and expansion to additional vital signs such as oxygen saturation and respiratory rate.

XII. CONCLUSION AND FUTURE WORK

This research has successfully developed a real-time heart health analysis system that offers a non-contact, accessible method for continuous cardiovascular monitoring. By integrating webcam-based video analysis with advanced signal processing techniques, the system provides accurate heart rate estimations without specialized equipment. Future work will focus on enhancing robustness in dynamic environments, expanding the system's capabilities to additional vital signs, and performing larger-scale clinical validation.

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